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Nguyen Vinh Truong [a], Lester W. Burgess [b] & Edward C.Y. Liew [c]

[a] School of Agriculture and Forestry, Hue University, Hue, Vietnam

[b] Faculty of Agriculture, Food and Natural Resources, University of Sydney, NSW, 2006, Australia

[c] Royal Botanic Gardens Sydney, Australia Botanic Gardens Trust, Mrs Macquaries Road, Sydney, NSW 2000, Australia

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Greenhouse and field evaluations of potassium phosphonate: the control of Phytophthora foot rot of black pepper in Vietnam

Nguyen Vinh Truong[a]*, Lester W. Burgess[b] and Edward C.Y. Liew[c]

[a]*School of Agriculture and Forestry, Hue University, Hue, Vietnam;* [b]*Faculty of Agriculture, Food and Natural Resources, University of Sydney, NSW 2006, Australia;* [c]*Royal Botanic Gardens Sydney, Australia Botanic Gardens Trust, Mrs Macquaries Road, Sydney, NSW 2000, Australia*

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Phytophthora foot rot of black pepper caused by *Phytophthora capsici* is a major disease of black pepper throughout production areas in Vietnam. The disease causes collar, foot and tap root rots and eventual death of the infected vine. Potassium phosphonate was evaluated for the control of this disease in greenhouse and field trials. In greenhouse trials three-month-old vines treated with phosphonate by soil drenching (10–20 g a.i./l) and then inoculated with *P. capsici* mycelium (2% v/v soil) had significantly less foot rot compared to vines grown in non-treated soil. In field trials mature vines were treated with phosphonate at 50–100 g a.i/pole soil drenching or 10 g a.i./l by root infusion. After 10 days root, stem and leaf specimens were removed for bioassay by inoculation with 5 ml of *P. capsici* zoospores suspension (10^6–10^8 spores/ml). Soil drenching with phosphonate inhibited the colonisation of pathogen on excised leaf, stem and root tissues, significantly more than phosphonate root infusion. Our study provides further evidence supporting the efficacy of potassium phosphonate in the management of black pepper foot rot caused by *P. capsici*. The excised leaf and stem bioassay used in this study is a rapid and useful technique for testing the efficacy of systemic fungicides in controlling this disease.

Keywords: bioassay; phosphonate; *Piper nigrum*; *Phytophthora capsici*

Introduction

Black pepper production in Vietnam is approximately 100,000 tons per annum, accounting for 60% of the global market (Viet Nam News 2007). Phytophthora foot rot of black pepper caused by *Phytophthora capsici* is a major constraint to black pepper production. Disease symptoms include sudden wilting of vines, defoliation, foot rot and root rot. Disease progress is very rapid; infected vines become dry and die in two to three weeks. The disease is a highly destructive and invariably leads to low productivity (Anandaraj 2000). In Vietnam, disease losses have been estimated to be more than US$18 million a year (Drenth and Sendall 2004). Although *P. capsici* was determined as a causal agent of Phytophthora foot rot, Tsao (1988) and other workers (Holliday 1980; Manohara et al. 2004) suspected that there might be

*Corresponding author. Email: nvinhtruong@huaf.edu.vn

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more than one species of *Phytophthora* attacking black pepper. A national disease survey in 2008 clarified the aetiology and confirmed that *P. capsici* was the main pathogen (Truong et al. 2008). The percentage of diseased black pepper poles ranged from 10.3% to 19.2% in different provinces. The highest percentage was 91.7% and some crops were abandoned and substituted with other crops.

Crop losses caused by *P. capsici* in Vietnam have been higher in comparison to other black pepper producing countries due to the lack of research on integrated disease management practises. The disease is favoured by poor drainage during the wet season and most of the commercial cultivars are susceptible. The most common black pepper growing soil in Vietnam is a basalt soil which has high moisture-retaining capacity and can remain wet for prolonged periods during the wet season. This situation favours the rapid formation of pathogen propagules, resulting in high disease incidence. Under such conditions, chemical controls are considered an important component of integrated disease management together with better drainage. Currently, disease control is mainly based on cultural practices used and fungicide use following the practices in other countries. More recently, a study of antagonistic bacteria conducted by Tran et al. (2008) reported biosurfactant-producing Pseudomonas in the rhizosphere of black pepper as a novel resource to control Phytophthora root rot of black pepper. However, this approach requires further experiments and field trials before being applied on a large scale in a developing country.

Potassium phosphonate, a mono- and di-potassium salt of phosphonic acid (H_3PO_3), is one of the phosphonate compounds that exist in aqueous solution in equilibrium with its tautomeric form (Dunhill 1990). The active part of this fungicide within plants is phosphonate (phosphite) or phosphonic acid which is the active ingredient working against the pathogen (Dunhill 1990; Grant et al. 1990). Potassium phosphonate is a truly systemic fungicide (Guest and Grant 1991) and is effective against plant diseases caused by many *Phytophthora* species (Guest et al. 1994; El-Hamalawi et al. 1995; Browne and Viveros 2005). Owing to the systemic nature of this chemical and its rapid metabolism within the plant, it has the potential to protect plant tissues and organs distant from the point of application (Nene and Thapliyal 1993). After application on the plant, the chemical is translocated upwards in the xylem and downwards in the phloem (Ouimette and Coffey 1990). The translocation of phosphonate to different parts of black pepper plant after 3 days application was demonstrated by using radioactive ^{32}P (Anil Kumar et al. 2009). Evidences from histological and biochemical studies show that phosphonate application increases levels of host defence to pathogen invasion (Jackson et al. 2000; Daniel et al. 2005; Daniel and Guest 2006). In plants treated with potassium phosphonate, the colonisation by the pathogen is severely restricted, the hyphae appear distorted and sporangial production is inhibited due to a rapid increase in the cytoplasmic activity. Development of cytoplasmic aggregates, release of superoxide, enhancement of phenolic compounds and hypersensitive cell death around the infected cell occur in the plant (Daniel and Guest 2006). Additionally, phosphonate has been found to be more effective when applied as a prophylactic prior to infection as this systemic fungicide is generally less effective as a curative agent (Marks and Smith 1992). The efficacy of phosphonate *in planta* may differ depending on the interaction between the host species and the Phytophthora strain (Guest and Grant 1991) and the longevity of fungicide efficacy can vary with plant species as well (Wilkinson et al. 2001).

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The efficacy of fosetyl-Al (aluminium tris-ethyl phosphonate), a ''sister'' phosphonate compound with potassium phosphonate, for the control of the Phytophthora diseases of chilli caused by *P. capsici* has been documented (Matheron and Porchas 2000). The usage of fosetyl-Al for Phytophthora foot rot of black pepper was also reported (Manohara et al. 2004) but the study of potassium phosphonate for control of this disease was limited. Potassium phosphonate was introduced in Vietnam in 1998 (Burgess, unpublished data). Some trials were conducted on farmer's orchards for the control of Phytophthora diseases of lychee (Burgess, unpublished data), durian (Diczbalis et al. 2004) and black pepper (Truong 2004). However, there is insufficient published evidence that demonstrates potassium phosphonate to be effective against foot rot disease of black pepper caused by *P. capsici*. Wong (2004) investigated the root infusion technique first used in Sarawak, Malaysia. Preliminary results indicated a reduction of disease spread and a prolonged life span of black pepper vines in infected farms. The aim of this study was to evaluate the optimum dosage and method of application of potassium phosphonate for the control of black pepper foot rot caused by *P. capsici* in greenhouse and field conditions.

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Materials and methods

Greenhouse trials

The optimal inoculum density for evaluation of fungicide efficacy and the optimal dosage and method of potassium phosphonate application for the control of black pepper foot rot caused by *P. capsici* were investigated in the greenhouse.

Experimental design

Three pot trials were conducted in the greenhouse. Each experimental unit comprised one pot with one black pepper vine in each pot. Free-draining plastic pots, 20 cm in height and 15 cm in diameter, were used in all experiments. Each experiment had one factor with 30 replications. The experiments were analysed in a randomised complete block design. In the first experiment, plants were inoculated with different inoculum levels of *P. capsici*. In the second and third experiment, plants were treated with different phosphonate concentrations after inoculation with *P capsici*.

Pathogen isolates and inoculum preparation

Twenty isolates of *P. capsici* obtained from various black pepper growing regions during a recent disease survey were selected for this study. These were maintained on potato carrot agar plugs in sterile deionised water at room temperature (Smith 2002). Their pathogenicity to black pepper had been verified by inoculation on leaves and young black pepper vines. Cultures were grown on potato carrot agar (PCA) (Dhingra and Sinclair 1995) at 22 $\pm$ 2°C for 5–6 days and used in inoculating sand-millet substrate for inoculum production. River sand was sieved using a 3-mm mesh, washed in running tap water for 10 min, then dried in the oven at 150°C for 3 h. The millet seed and sand was mixed at a ratio of 1:2, and 600 ml of the medium was poured into 2-l Erlenmeyer flasks together with 400 ml of deionised water. The flasks were wrapped with aluminium foil before autoclaving at 121°C and 100 kPa for 30 min and the

medium was cooled to room temperature before inoculation. The medium was inoculated with agar blocks from 5-day-old cultures of *P. capsici* on potato dextrose agar (PDA) (Dhingra and Sinclair 1995), one isolate per flask. The inoculated flasks were shaken and incubated for two weeks at room temperature (22 $\pm$ 2°C), by which time they were fully colonised by the pathogen. The inoculum from all the flasks was evenly mixed in a sterilised container before inoculating into the pots. The inoculum consisting of colonised sand-millet seed was harvested and then mixed thoroughly before being incorporated into the pots containing black pepper vines.

Plant preparation

Basalt soil, a red cracking clay soil formed from volcanic rocks, was obtained from areas without diseased black pepper vines, dried and sieved before use. No fertiliser was added to the soil during the experiment. Black pepper vines of the commercial cultivar Vinh Linh were grown from cuttings obtained from healthy mature plants. After three months, young healthy black pepper vines approximately 40–50 cm in height were used for the trials.

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Experiment 1

The objective of this experiment was the investigation of *P. capsici* inoculum concentration on the development of Phytophthora foot rot of black pepper vines. A layer of soil was removed from each pot; then various amounts of the sand-millet seed inoculum were spread onto the surface of the remaining soil at proportions of 0, 1, 5, 10 or 20% (v/v soil). The inoculum was mixed throughout the soil in each pot; then the soil originally removed from each pot was replaced. The surface of each pot was covered with aluminium foil and each pot placed on a separate shallow plastic saucer to provide continuous moisture, as well as to prevent cross-contamination. The pots were watered daily with tap water until saturation throughout the experiment.

Experiment 2

The objective of this experiment was to investigate the phosphonate dosage of soil drenches on Phytophthora foot rot of black pepper vines. Plant preparation, inoculum production and plant inoculation techniques were as described in Experiment 1. Potassium phosphonate used in this experiment was Agrifos 400 (400 g/l phosphorous acid present as mono-di potassium phosphate, Agrichem Manufacturing Pty Ltd). The chemical was diluted with 0, 1, 5, 10 and 20 g a.i./l of tap water. A 100-ml aliquot of chemical solution was applied to each pot by soil drenching. Control pots were treated with 100 ml of tap water. Ten days after the phosphonate treatment, inoculum of sand-millet seed was added at a rate of 2% (v/v), the optimal concentration described in Experiment 1. The plants were maintained and watered as in Experiment 1.

Experiment 3

The objective of this experiment was to investigate the phosphonate applied as a foliar sprayer on the development of Phytophthora foot rot of black pepper vines. Plant

preparation, inoculum production and plant inoculation techniques were as described in Experiment 1. Phosphonate was applied to the foliage with a hand spray until run-off without using any surfactant. Chemical application rates were 0, 1, 5, 10 and 20 g a.i./l and control pots were sprayed with tap water. Ten days after phosphonate treatment, the plants were inoculated and watered as described in Experiment 1.

Disease assessment

The black pepper vines were monitored daily for symptom development in all three experiments. The number of dead and healthy plants in each treatment was recorded every second day after phosphonate application and inoculation until the end of the experiment. The pathogen was re-isolated from plants showing disease symptoms on Phytophthora selective medium (PSM) (Singleton et al. 1992) to verify the causal agent as *P. capsici*.

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Field trials

These trials involved the determination of the optimal dosage and method of potassium phosphonate application for the protection of black pepper vines against *P. capsici* in the field.

Plant selection

Black pepper vines (7–8-years old) were selected from a farm with red basalt soil in Cam Nghia Commune, Cam Lo District, Quang Tri Province, Vietnam.

Trial 1

The objective of this trial was the investigation of phosphonate field soil drenching on the colonisation of *P. capsici* assessed by leaf, stem and root bioassays. The experiment was based on a randomised complete block design with one replication. There were three treatments, three black pepper rows per treatment. Each row consists of 10 poles, each with three to four vines. Black pepper vines in the treatments were treated with phosphonate at 0, 50 or 100 g a.i./pole. The phosphonate solution was diluted with well water at concentrations of 0, 50 and 100 g a.i./l and a volume of 10 l of chemical solution was applied as a drench around each black pepper poles. Different plant parts were inoculated with zoospores of *P. capsici* in the laboratory 10 days after the chemical treatment. Leaves were removed at 0.5, 1.0 and 1.5 m from the ground level and 50 leaves were collected at each distance for each treatment. Green stems were removed 0.5 m from the ground level and 50 stems were collected for each treatment. Five to ten pieces of primary roots, about 4–8 mm in diameter and 50–100 cm in length, were removed and tested for each treatment. Samples were placed in plastic bags at temperature about 20°C and transported to the laboratory in an insulated esky.

Trial 2

The objective of this trial was the investigation of phosphonate root infusion in the field on the colonisation of *P. capsici* assessed by leaf and stem bioassays. The

experiment had two treatments in a row with 10 experimental black pepper poles and 10 control poles all at one-pole intervals. Roots of vines in each treatment were treated with phosphonate at 0 or 1 g a.i./l by the root infusion technique (Wong 2004). Healthy primary roots of 6–10 mm in diameter regularly distributed around the black pepper pole were chosen. Soil was dug out carefully from around the roots with a hand trowel so as to reveal the roots. Secondary roots and rootlets on the primary roots were removed and the primary roots were cleaned with water. The roots were then cut with a sharp knife and inserted immediately into a plastic bottle containing 100 ml of phosphonate solution. The bottles were kept in place at an angle for 1 day and shaded with straw. The bottles were removed the following day and the soil was replaced. Green stems and leaves were removed for laboratory testing 10 days after the phosphonate treatment as described in Trial 1.

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Inoculum preparation for bioassay

P. capsici zoospores were produced under aseptic conditions following a modification of the procedures reported by Ristaino (1990). PCA agar plugs maintained in sterile deionised water were transferred onto PCA (Dhingra and Sinclair 1995) and incubated in the dark at 22 $\pm$ 2°C. Sub-cultures were obtained from the edge of a 4-day-old colony and plated onto carrot agar (CA) (Erwin and Ribeiro 1996). Plates were incubated in the dark for 3 days at 22 $\pm$ 2°C before being transferred to continuous fluorescent light for a further 3 days to induce sporangium and zoospore formation. After adding 10 ml of deionised water to each plate, the plates were placed in a fridge (4°C) for 1 h before being returned to fluorescent light at 22 $\pm$ 2°C for a further 1 h until zoospores were released abundantly. The zoospore suspension from each plate was transferred to a 15-ml test-tube and left undisturbed for 5 min to allow mycelial fragments, sporangia and agar debris to settle. The upper 4–5 ml of suspension containing free-swimming zoospores was then transferred to a 50-ml beaker. Zoospore concentration was estimated with a haemocytometer and adjusted to 10^6–10^8 spores/ml.

Inoculation

Detached plant parts were inoculated *in vitro* using the methods described by Dhingra and Sinclair (1995). Leaf, stem and root samples were rinsed with tap water, dried on tissue and swabbed with 70% ethanol. Stem and root samples were cut into sections 10 cm in length. The cut surface of petiole, stem and root sections was sealed with paraffin wax to minimise water loss. Thirty leaves, stems, or root sections each were placed in a metal tray in three rows on layers of moist tissue. Stem and root sections and leaves were pricked with a sterile needle and inoculated with 5 μl of zoospore suspension. Uninoculated samples were similarly injured but inoculated with sterilised deionised water. After inoculation, all trays were covered with clear film wrap and incubated in the dark at room temperature (22 $\pm$ 2°C).

Disease assessment

Length of lesion on the leaves and stems were measured daily for 3 days after inoculation. Length of lesion on root sections was assessed below the epidermis on the fifth day after inoculation. The presence of the pathogen in diseased leaf, stem

and root tissues inoculated with *P. capsici* was verified by plating subsections of the tissue onto PSM (Singleton et al. 1992).

Statistical analysis

In the greenhouse experiments, the number of healthy plants was analysed using a logistic regression analysis in GenStat v. 8.1 (Rothamsted Experimental Station). In field Trial 1, the lesion lengths of excised leaves and stems were used to calculate the area under disease progress curve (AUDPC) and rate of infection (r). AUDPC and r is an index of the amount of disease developed on the host and the slope of the regression line of disease progress data (Singh and Rao 1989). The AUDPC was calculated using GraphPad Prism v. 4 (GraphPad software Inc., San Diego, CA). The rate of infection was calculated and analysed by ANCOVA (Zar 1999) (GraphPad Prism v. 4). The lesion lengths of excised roots were analysed using Tukey's Multiple Comparison Test ($P = 0.05$) (GraphPad Prism v. 4). In field Trial 2, the lesion lengths of excised leaves and stems were analysed using the Mann-Whitney procedure (GenStat v. 8.1). Sections inoculated with sterile deionised water without showing disease symptoms and with no pathogen re-isolated were not included in the statistical analysis.

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Results

Greenhouse trials

Experiment 1

The first wilt symptoms on the black pepper vines appeared 2 days after inoculation and the first vine death due to foot and root rot was recorded on the third day. There were significant differences in the number of healthy vines inoculated with different inoculum concentrations 7 days after inoculation (data not shown). More vine deaths were observed at high inoculum levels and all vines were inoculated at concentrations of 10% and 20% (v/v) had died by the end of the experiment 29 days after inoculation. A strong negative association between the number of healthy vines and inoculum concentration of *P. capsici* was obtained from the logistic regression analysis of the data on the 29th day after inoculation (Table 1). Approximately 64% of the vines were diseased at 1% inoculum level two weeks after inoculation. *P. capsici* was recovered from all vines showing disease symptoms. The results indicated that inoculum concentrations from 1% to 5% (v/v) were able to cause 50–80% of diseased vines at day 10 after inoculation (data not shown). Inoculum concentration of 2% (v/v) was chosen for Experiments 2 and 3, as this inoculum level provided an amount of diseased vines deemed reliable for the evaluation of fungicide effect.

Experiment 2

Generally, the number of healthy vines treated with phosphonate was greater than phosphonate non-treated vines. The logistic regression analysis indicated a significant positive correlation between the number of healthy vines and phosphonate concentrations at 10 and 20 g a.i./l ($P \leq 0.05$) (Table 1). The results indicate that a concentration of fungicide from 10 to 20 g a.i./l effectively protected

Table 1. Correlation[a] between inoculum concentration of *Phytophthora. capsici*, phosphonate concentration and the number of healthy black pepper vines on the twenty-nin29th day after inoculation, in the greenhouse.

Parameters[b]	Estimate[c]	s.e.[d]	t(*)[e]	t pr.[f]
Inoculum				
Control[g]	1.60	0.49	3.29	0.001
1% (v/v)	−2.30	0.62	−3.69	<.001
5% (v/v)	−4.98	1.23	−4.41	<.001
10% (v/v)	−12.20	21.80	−0.56	0.577
20% (v/v)	−12.20	21.80	−0.56	0.577
Soil drenching				
Control	−1.01	0.41	−2.46	0.014
Phosphite 1 g (a.i./l)	0.74	0.55	1.35	0.178
Phosphite 5 g (a.i./l)	0.74	0.55	1.35	0.178
Phosphite 10 g (a.i./l)	1.56	0.55	2.79	0.005
Phosphite 20 g (a.i./l)	1.56	0.55	2.79	0.005
Foliar spray				
Control	0.00	0.36	0.00	1.000
Phosphite 1 g (a.i./l)	−0.69	0.53	0.53	0.192
Phosphite 5 g (a.i./l)	−0.55	0.53	−1.04	0.299
Phosphite 10 g (a.i./l)	−0.40	0.52	−0.78	0.437
Phosphite 20 g (a.i./l)	−1.19	0.56	−2.11	0.035

[a]Correlation was analysed using logistic regression analysis.
[b]Parameters for concentrations were differences compared with 0% (v/v) inoculum or 0 g phosphonate.
[c]Estimates of the intercept and slope parameters in the model.
[d]Standard error of slope and intercept.
[e]Percentage of a *t*-distribution with $(n-2)$ degrees of freedom (the number of degrees of freedom on which the estimate s^2 was based).
[f]The *t* probabilities for the parameter estimates.
[e]Controls were treated with sand-millet seed, water drench or water spray.

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the black pepper vines from foot rot caused by *P. capsici* for 29 days. *P. capsici* was recovered from all vines showing symptoms.

Experiment 3

There was no significant reduction in the number of healthy vines with higher levels of phosphonate concentrations. No association between the number of healthy vines and phosphonate concentration was found in the logistic regression analysis (Table 1). Phytotoxicity symptoms were very obvious on vines sprayed with phosphonate at concentrations of 5 g a.i./l and above (Figure 1). These symptoms were observed mainly as a burning of shoots and leaves, sometimes even the entire stem. The symptoms mostly started after 2 days treatment from the edge and tip of the leaf, spreading to the inner part of the lamina until the entire leaf was burnt and shed. Vines became desiccated and were often dead within one to two weeks. Vine death due to phytotoxicity was easily distinguished from that caused by the disease. Vines affected by phytotoxcity were burnt and turned rusty brown in colour, whereas diseased vines were wilted with leaves remaining pale green. The wilt symptoms on diseased vines were usually abrupt and the pale green leaves were seldom shed.



Figure 1. Symptoms of diseased young black pepper vine with leaves withered but intact (A) 3 days after inoculation and phytotoxicity affected vine with burnt leaves that were mainly shed (B) 3 days after phosphonate treatment at 10 g a.i./l. Scale bars = 10 cm.

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Field trials

Trial 1

The AUDPC of leaves and stem removed from the phosphonate-treated vines was significantly different from those with untreated vines (Table 2) but there was no significant difference in AUDPC between phosphonate concentrations 50 and 100 g a.i./pole. ANCOVA analysis showed that the *r* values of the excised leaves from black pepper vines drenched with either 50 or 100 g a.i/pole phosphonate solutions were significantly different as compared to those of untreated vines ($P \leq 0.05$) but there was no significant difference between the two concentrations (Table 2). For excised stems, the *r* values of the lesion lengths from vines drenched with 0, 50 or 100 g a.i/pole phosphonate solution were significantly different. The lowest *r* value was that of the 100 g a.i./pole treatment. For excised roots, the lesion lengths below the epidermis of the sections from vines treated with phosphonate solution at different concentrations were significantly ($P \leq 0.05$) different (Figure 2). In general, phosphonate treatment significantly ($P \leq 0.05$) decreased the colonisation of *P. capsici* on the excised roots but the 100 g a.i/pole treatment had the highest effect in suppressing disease development.

The uninoculated control did not show any symptoms, verifying that the disease lesions observed in the inoculated treatments were caused by the *P. capsici* inoculum. Not all plant sections inoculated with *P. capsici* showed external disease symptoms, although a characteristic lesion developed on most individuals. It was observed that the development of lesions on the excised leaf, stem and root was different. The

necrotic area on the leaf had light and dark-brown concentric rings with fimbriate margins, while necrosis extended both upwards and downwards from the point of inoculation on the stem and root. Additionally, the root lesion exuded a dark fluid. The leaf lesion was apparent 1 day after inoculation, but the lesion appeared later for stem (2 days) and root (3 days) inoculation. These lesions were observed on samples removed from both phosphonate-treated and untreated vines. However,

Table 2. Area under disease progress curve (AUDPC)[a] and the rate of infection (r)[b] of lesion development after zoospore inoculation of *Phytophthora. capsici* on excised black pepper leaves and stems from field-grown plants 10 days after soil drenching with phosphonate concentration at 0, 50 or 100 g a.i./pole.

Plant tissue sample (distance above ground)	Phosphonate concentration (g a.i./pole)					
	0		50		100	
	AUDPC	r	AUDPC	r	AUDPC	r
Leaf 1 (0.5 m)	21.8	8.6 ± 0.4	13.4	5.8 ± 0.4	14.7	5.8 ± 0.2
Leaf 2 (1.0 m)	21.3	8.3 ± 0.4	16.1	6.2 ± 0.2	17.2	6.6 ± 0.3
Leaf 3 (1.5 m)	22.9	8.8 ± 0.3	16.5	6.1 ± 0.3	16.2	6.3 ± 0.2
Stem	5.9	3.6 ± 0.7	3.2	2.2 ± 1.0	0.9	0.3 ± 0.4

[a]AUDPC was calculated using means of lesion lengths calculated from 30 replicates.
[b]r was calculated and analysed by ANCOVA ($P < 0.05$).

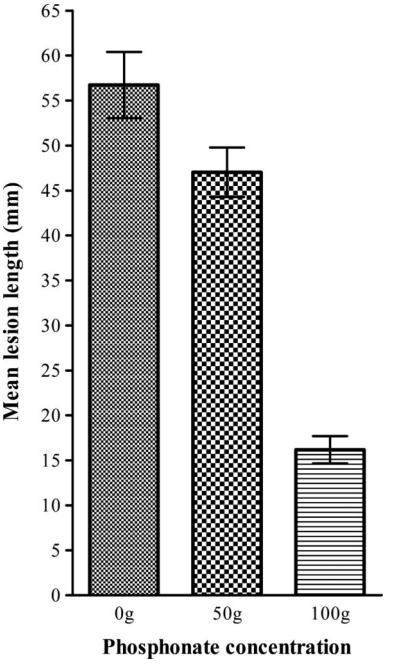


Figure 2. Mean lesion length (± standard error of the mean) 5 days after inoculation of excised roots removed from black pepper plants, treated with phosphonate at three rates (0, 50 and 100 g a.i/pole) by soil drenching and subsequently inoculated with *P. capsici* zoospores.

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necrotic areas were larger on vines that had not been treated with phosphonate. The visible lesion length on excised leaves, stems and roots increased after inoculation but the rate of development was significantly ($P \leq 0.05$) different. Potassium phosphonate, as indicated in the *in vitro* bioassays, significantly ($P \leq 0.05$) inhibited the activity of *P. capsici* on the excised leaves, stems and roots of black pepper from phosphonate-treated vines at either 50 or 100 g a.i./pole.

Trial 2

Significant differences in visible lesion length of excised stems between phosphonate-treated and untreated vines at a distance of 0.5 m from the ground were observed ($P < 0.001$) (Table 3). The visible lesion lengths of the sections from phosphonate-treated vines were shorter than those on untreated vines. Similarly, the rate of lesion development on excised leaves from phosphonate-treated vines 1 m from the ground was significantly less than those from untreated vines. However, there were no significant differences in the visible lesion lengths of the excised leaves removed from vines at distances of 0.5 and 1.5 m from the ground level between sections from phosphonate-treated and untreated vines.

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Discussion

Our results demonstrate that potassium phosphonate significantly reduced the activity of *P. capsici* in black pepper vines under greenhouse and field conditions. Two concentrations of potassium phosphonate (10 and 20 g a.i./l) applied by soil drenching in pots before inoculation effectively protected young black pepper vines from the pathogen. The results in our field experiments were also consistent with those obtained in the greenhouse trials. The excised leaves, stems and roots from freed plants drenched with 50 or 100 g a.i./pole phosphonate solution and subsequently inoculated with *P. capsici* zoospores were shown to be more resistant to disease development.

Phosphonate anions can act directly on the pathogen, alter the host response and change the host reaction from compatible to incompatible (Guest and Grant 1991).

Table 3. Effect of phosphonate applied as a root infusion on the subsequent development of *P. capsici* lesions on excised leaves and stems of black pepper 10 days after phosphonate (P) application.

Plant tissue sample	Mean (mm)[a]		U-value[b]	p-value[c]
	P-treated	Non-P-treated		
Leaf 1[d]	2.0	2.9	352.5	0.141
Leaf 2	1.4	4.0	147.0	<0.001[e]
Leaf 3	2.2	2.6	410.0	0.547
Stem	1.0	4.4	113.0	<0.001[e]

[a]Mean visible lesion length on black pepper leaves three 3 days after inoculation.
[b]The value of U statistics. U as calculated by the Mann-Whitney procedure.
[c]The probability density of U.
[d]Leaf 1, leaf 2 and leaf 3 were excised from black pepper plants at respective distances of 0.5, 1 and 1.5 m from the ground level.
[e]Significant at $\alpha = 0.05$.

The suppression of the growth rate, the ability of hyphal lysis and the disruption of the biosynthetic pathway of polyunsaturated long chain fatty acids of *P. capsici* by phosphonates *in vitro* have been documented (Soulie et al. 1995; Wong 2006). In addition, phosphonate-treated plants appear to be able to generate an antimicrobial environment more effectively than those not treated with the chemical, by disrupting pathogen metabolism and stimulating their own defence mechanisms (Guest and Grant 1991; Daniel et al. 2005; Daniel and Guest 2006). Plants treated with phosphonate have been demonstrated to cause stress in the pathogen and prompt a host defence response. Phosphonate-elicited defence response in plants following challenge by Phytophthora pathogen is multifaceted and coordinated and involves rapid cytoplasmic aggregation, nuclear migration, superoxide releases, hypersensitive cell death and the accumulation of phenolic compound. Phytoalexins and their derivatives in the infected and neighbouring cells were produced more rapidly after challenge with a potential pathogen than those not treated (Daniel et al. 2005; Daniel and Guest 2006). Disease development on the excised black pepper leaves was different from those on excised stems and roots. This may be due to a differential response of the plant's defence mechanisms. Differences in the level of colonisation of different tissues by the pathogen have been reported in other plant–pathogen interactions (Mercado-Blanco et al. 2003; Silvar et al. 2005). *P. capsici*'s ability to infect pepper leaf tissue more readily than stem and root tissue is often used to bait and isolate *P. capsici* from soil (Kueh and Khew 1982). The current results are in agreement with the report of Veena and Sarma (2000) where potassium phosphonate effectively protected black pepper plant against Phytophthora foot rot in India.

Of the methods of application tested in our experiment, soil drenching with potassium phosphonate was more effective than foliar sprayer or root infusion for the protection of black pepper vines against the colonisation of *P. capsici*. In addition, phytotoxicity symptoms appeared when phosphonate was applied by foliar sprayer at concentrations of 5 g a.i./l or above. Although potassium phosphonate is generally considered to have low phytotoxicity (Guest and Grant 1991), phytotoxicity has nevertheless been reported in crops and plants (Pilbeam et al. 2000; Vawdrey et al. 2004). Phosphonate caused foliar burning and defoliation at high application concentrations in mandarin (Walker 1989), carrot (Walker 1991) and *Pinus radiata* (Ali et al. 1999). According to Walker (1989), foliar necrosis may result from an osmotic stress created by a high concentration of soluble salts.

The phosphonate root infusion technique developed by Wong (2004) was designed to increase the level of phosphonate uptake in root and vine tissues by preventing run off and leaching. This technique helps to ensure no wastage of chemical and worker safety, thus reducing cost, chemical hazard and environmental pollution. Although this technique was reported to have been applied in Malaysia, there were not sufficient data to prove its effectiveness. In the current study potassium phosphonate applied by root infusion was significantly less effective than soil drenching in suppressing the activity of pathogen on excised leaves, stems and roots. One factor that could partially account for these differences is that the level of phosphonate absorption by the primary roots of black pepper plant may be less than that of root hairs involved in chemical uptake by soil drenching. In addition, black pepper plants are often grown in clumps of three to five vines per pole in the field. This leads to the difficulty of tracing the exact infused roots to upper plant parts for bioassays. Further investigation is required to evaluate the efficacy of potassium phosphonate applied by root infusion for the control of Phytophthora foot rot.

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The bioassay of using excised leaves, stems and roots described in this study proved to be a reliable method for assessing the systemic antimicrobial activity in black pepper. The parts of the black pepper vines treated with potassium phosphonate and used in the bioassays consistently showed suppression of the activity of pathogen. However, using excised leaves and stems was easier than excised roots. Excised leaves and stems of black pepper have previously been used to screen for disease resistance (Turner 1971), evaluate antagonistic bacteria (Anith et al. 2003) and fungicides (Turner 1969). Such bioassays are used to expedite the identification of disease-resistant cultivars or suitable chemicals for disease control. Our study shows that the excised leaf and stem bioassay, being efficient in terms of space, time and cost, is useful for preliminary evaluation of the systemic activity of chemicals against Phytophthora foot rot of black pepper. In addition, systemic fungicides can be evaluated quickly under a controlled laboratory environment that is experimentally more favourable than field conditions. Although such bioassays have limitations in simulating natural infection processes in the field, they are nevertheless useful in providing a good indication of fungicide efficacies before warranting further testing in the field.

Our results indicate that potassium phosphonate has considerable potential for use in the management of Phytophthora foot rot of black pepper in Vietnam. Although phosphonate resistance in *Phytophthora* species has not been detected in any crop after 25 years of constant use in various countries, the capacity to develop phosphonate tolerance in *Phytophthora* species has been reported in the laboratory (Bower and Coffey 1985; Dolan and Coffey 1988). One report claimed that *P. capsici* developed high tolerance to both phosphonic acid and fosetyl-Al *in vitro* (Bower and Coffey 1985). Hence, the potential risk of this occurring cannot be ignored under natural field conditions. Successful sustainable management of Phytophthora foot rot in black pepper should involve the implementation of integrated disease management strategies with chemical control as only one component, as exemplified by the management of Phytophthora blight of bell pepper suggested by Ristaino and Johnston (1999). In terms of economic and environmental considerations, genetic resistance would still be the best management strategy of Phytophthora diseases when integrated with chemical and cultural practices.

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